

Influence of the combined additions Mn, Cr or Ni on the formation of dispersoids and mechanical properties of Al–Mg–Si–Cu alloys

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ABSTRACT

The effect of individual and combined additions of different transition metals (TMs), Mn, Cr, and Ni, on the precipitation of dispersoids and the final mechanical properties of Al–Mg–Si–Cu alloys was systematically studied by scanning electron microscopy, transmission electron microscopy and tensile testing. The various TM additions examined are found to induce notably different dispersoid precipitation behaviors. The combined addition of Mn, Cr produces the highest number density of dispersoids among the three TM-added alloys examined, due to the incorporation of Cr promoting the precipitation of the α -Al(Mn, Cr, Fe)Si dispersoid. By contrast, the combined addition of Mn, Ni slightly suppresses the precipitation of the α -Al(Mn, Ni, Fe)Si dispersoid compared with individual Mn addition. The presence of a fine dispersoid microstructure can moderately increase the strength, and strongly increase the elongation, especially for the case of the combined Mn, Cr added alloy, where the elongation is increased almost threefold compared with the base alloy. An excellent strength-elongation combination, with an ultimate tensile strength of 421.9 ± 2.2 MPa and an elongation of 18.2 ± 3.6 , is achieved for this alloy. This property is superior to most reported 6xxx alloys. The main reasons for this result are the presence of a high number density of dispersoids promoting grain refinement, providing high Orowan hardening, and reducing stress concentration on the grain boundary. The findings of this work are expected to provide guidance for developing new Al–Mg–Si–Cu alloys with excellent comprehensive mechanical properties.

1. Introduction

6xxx Al–Mg–Si alloys are among the most important extruded and rolled aluminum alloys, widely used in automotive, aerospace, and construction industries due to their medium to high strength, good formability, and excellent corrosion resistance. The former of these properties emerges owing to the formation of a high number density of nano-sized, (semi-)coherent precipitates, formed during aging treatment, that obstruct the movement of dislocations and strengthen the alloy [1–3]. However, the interaction between dislocations and precipitates generally leads to the so-called ‘strength-ductility trade-off’ (increasing strength causing ductility loss), which is a long-standing

dilemma not only for the age hardenable Al alloys, but also for most metals [4,5]. For the peak aged Al–Mg–Si alloys, the yield strength is mainly ranging from 210 to 360 MPa. However, the elongation has significantly decreased once reaching peak-aged condition, which limits the material application [6]. Therefore, simultaneously promoting the strength and elongation of the Al–Mg–Si alloys should have significant implication to the application potential for these materials.

The dispersoids formed as a result of transition metal (TM = Mn, Cr, Ni, Co) addition to the Al–Mg–Si alloys represent a potentially feasible approach for tailoring the strength-ductility trade-off. Normally, these dispersoids are introduced to control recrystallization during deformation and annealing. Following a small addition of TMs, TM-containing

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dispersoids are formed upon high temperature homogenization (400–550 °C) [7–10]. Due to their excellent thermal stability and high number density, these particles can strongly retard the recovery and recrystallization processes, and may even act as heterogeneous nucleation sites for the precipitation, greatly affecting the alloy mechanical properties [11,12]. Recently, Arani et al. [13] reported that the presence of Mn dispersoids can reduce stress concentrations on the grain boundary and thus decrease the propensity for intergranular fracture. The strain to fracture is approximately doubled after Mn addition [14]. This result shows that the TM-containing dispersoids have great potential for increasing the comprehensive mechanical properties of the Al–Mg–Si alloys.

The effect of TM addition on alloy microstructure and mechanical properties has been widely studied in previous works. Among the various TMs, Mn is the most attracting element used in Al–Mg–Si alloys. Mn addition can promote the precipitation of α -Al(Mn,Fe)Si dispersoids during homogenization. These particles adopt a simple cubic/body-centered cubic (sc/bcc) structure and are partially coherent with the Al matrix [15]. Li et al. [16] reported that 6082 alloy containing 0.5 % Mn exhibits the highest recrystallization resistance among the studied alloys due to a fine dispersoid distribution and rather narrow dispersoid-free zones (DFZs). Qian et al. [12] showed that the formation of α -Al(Mn,Fe)Si dispersoids facilitates the retardation of recrystallization due to the pinning effect of these particles on dislocation movement and sub-grain boundary migration. Cr is another TM commonly used in Al–Mg–Si alloys. Lodgaard et al. [17] studied the precipitation of a Cr-containing dispersoid during homogenization. They found that small α -Al(CrFe)Si dispersoids formed at the surface of β -AlFeSi particles, with the former particles growing and the latter dissolving upon heating to higher temperature. An et al. [18] reported that Cr addition to an Al–Si–Mg–Cu alloy produced a stronger hardening response due to the precipitation of α -Al(CrFe)Si dispersoids. Recently, Yuan et al. [19] studied the effect of Ni micro-alloying on the formability and bendability of Al–Mg–Si alloys. It was revealed that Ni addition could promote a uniform distribution of Ni-containing Fe-rich dispersoids, and thus improve the formability and bendability of the alloy. Wang et al. [20] studied the effect of different TMs on the age hardening of Al–Mg–Si alloys. Compared with the base alloy, the peak hardnesses are higher for the Co- or Ni-added alloys, lower for the Mn-added alloy, and unchanged for the Fe- or Cr-added alloys. This outcome is mainly due to TMs forming dispersoids with different amounts of Si, giving rise to different precipitation hardening responses as a result of the different levels of Si consumed from the base alloy. By comparison, the effect of dispersoids on the strength-ductility trade-off for the alloy has received less attention.

The above investigations have confirmed the beneficial effect of dispersoids on selected alloy properties. How to promote the precipitation of a high number density of dispersoids however remains a critical issue. Recently, increased attention has been devoted to the combined effects of different TMs on the precipitation of dispersoids in Al–Mg–Si alloys. Liu et al. [21] studied the effect of a varying Mn/Cr ratio on the recrystallization and mechanical properties of the 6xxx alloys. A Mn/Cr ratio of 3.5 contributed to the maximum number density of fine dispersoids, greatly increasing alloy recrystallization resistance and dispersion strengthening. Raul et al. [22] reported that combined additions of Cr, Zr and Mn could greatly increase the creep resistance of an Al–Si–Mg–Cu alloy at 300–350 °C due to the formation of a high number density of fine α -Al(Cr,Mn, Fe)Si and ϵ -Al-Si-Zr phases. However, the detailed mechanism of the individual and combined TM additions on the precipitation of the dispersoids is still not fully understood: The evolution of these particles during homogenization, and their effect on the final microstructure and properties in combined TM-added alloys remain elusive. Especially, the combined additions of Mn, Cr and Mn, Ni on the precipitation of dispersoids are far away from being understood [21]. Therefore, thoroughly probing the effect of combined TM addition on the dispersoid formation and mechanical properties of Al–Mg–Si–Cu

alloys is an issue of great scientific and technological implication to the wider application of these alloys.

In the present work, four different Al–Mg–Si–Cu alloys with different TM additions were examined: TM-free base alloy, individual Mn-added alloy, combined Mn-, Cr-added alloy and combined Mn-, Ni-added alloy. The effect of combined TM additions on the precipitation of dispersoids during homogenization and on the alloy mechanical properties after final aging was systematically studied. The combined effects of the combined TM addition are highlighted.

2. Experimental procedure

Four Al–Mg–Si–Cu alloys with different additions of TMs (TM = Mn, Cr, Ni) were prepared in this study. The alloy compositions are listed in Table 1. Alloy 1 is the base alloy, alloy 2 is a 0.5Mn-added alloy, while alloys 3 and 4 are combined 0.5Mn-0.2Cr- and 0.5Mn-0.2Ni-added alloys, respectively. All numbers in these descriptions denote wt.%. The alloys were cast into a slab ingot from high-purity Al (99.9 %), high-purity Mg (99.9 %), high-purity Zn (99.9 %), Al-10 wt% Si, and Al-49.5 wt% Cu master alloys using an induction furnace. Mn, Cr and Ni were added in the form of Al-10 wt%Mn, Al-10 wt%Cr and Al-10 wt%Ni master alloys. The chemical compositions of the alloys were measured by optical emission spectroscopy. The investigated alloys were homogenized at 560 °C for 6h and then air cooled to room temperature. Two different heating rates were applied for the homogenization: ‘rapid heating’ (carried out in a muffle furnace once the temperature reached 560 °C), and ‘slow heating’ (50 °C/h to the desired final temperature). Afterwards, the samples were undergone hot rolling at 510 °C from 20 mm to 6 mm by 6 passes, then cold rolling to 1 mm thick sheet. The cold-rolled sheets were solution treated at 560 °C for 30min, water-quenched to room temperature and immediately artificially aged at 180 °C for different durations.

The Vickers hardness of the alloys was tested using a Q10 CHD Master + type microhardness instrument with a holding time of 10s and a force of 500g. Ten indentations were made for the sake of calculating the average hardness of the alloys. Tensile testing of the alloys was performed on an MTS C43.504 type testing machine at a stretching rate of 1 mm/min. To ensure accuracy in the experiments, three parallel samples were tested for each condition. Information on the distribution and evolution of the primary phases and dispersoids was acquired by scanning electron microscopy (SEM, TESCAN S8000 GMH). The microstructure and composition of the dispersoids were characterized by transmission electron microscopy (TEM, Thermal Fish Tecnai G2 F20) operated at 200 KV. The statistical average sizes, number densities and volume fractions of the dispersoids were obtained by Image J software, with at least ten images being studied to ensure accuracy in the analysis. The dispersoid volume fraction V was calculated using the following equation:

$$V = A_A \frac{K\bar{D}}{K\bar{D} + t} (1 - A_{DFZ})$$

Where A_A is the area fraction of the dispersoids, $\bar{D}$ is the average equivalent diameter of the dispersoids in the TEM images, A_{DFZ} is the area fraction of the dispersoid free zones, t is the TEM foil thickness. Sample thicknesses were measured by convergent beam electron diffraction (CBED). All TEM specimens were prepared by electro-polishing at –25 °C, using an electrolyte consisting of 25 vol% HNO₃ and 75 vol% CH₃OH.

Table 1
Chemical compositions of the four investigated alloys (wt.%).

	Mg	Si	Cu	Zn	Fe	Mn	Cr	Ni	Al
1 #	1.13	0.9	0.5	0.21	0.05	–	–	–	Bal.
2 #	1.11	0.92	0.49	0.23	0.05	0.5	–	–	Bal.
3 #	1.14	0.9	0.47	0.2	0.06	0.47	0.18	–	Bal.
4 #	1.12	0.89	0.5	0.21	0.05	0.5	–	0.19	Bal.

3. Results

3.1. As-cast microstructures

Fig. 1 shows back scattered electron (BSE) images of the four as-cast alloys. All alloys exhibit a clear eutectic structure in this condition, consisting of α -Al dendrites, dark primary Mg_2Si , and bright Fe-rich intermetallics. Almost all the primary phases are continuously or discontinuously distributed along the grain boundaries. Enlarged SEM images and corresponding energy-dispersive X-ray mappings of the primary phases are presented in Fig. 2. The SEM-EDS point analysis is presented in Table 2 to identify the type of the primary phases. For the base alloy 1, the bright Fe-rich intermetallics can be identified as α -AlFeSi phase due to the script-like morphology. The primary Mg_2Si phase can also be identified, based on the enrichment with Mg and Si, and the rather low BSE intensity. For alloy 2 (3), Mn (Mn, Cr) are incorporated in the Fe-rich intermetallics, forming α -Al(Fe,Mn)Si (α -Al(Fe,Mn,Cr)Si) phase. In addition to the Mg_2Si phase, a low density of bright spherical particles is formed in the grain interior. These particles are inferred to be the Q- $\text{Al}_5\text{Cu}_2\text{Mg}_8\text{Si}_5$ phase, based on the enrichment with Mg, Si and Cu (marked with a yellow arrow in Fig. 2(b)). It should be noted that the Q phase is formed in all alloys, but it is not present in Fig. 2(a, c, d) due to its low number density. For alloy 4, in addition to the formation of α -Al(Fe,Mn,Ni)Si and Mg_2Si phases, some eutectic AlCuNi phase is observed. Polarized light optical microscope (PLOM) maps of the four as cast alloys are shown in supplementary materials Fig. S1. The grain sizes of the four alloys are measured as $42.1 \pm 3.6 \mu\text{m}$, $47.2 \pm 5.4 \mu\text{m}$, $79.2 \pm 5.6 \mu\text{m}$ and $46.1 \pm 4.3 \mu\text{m}$ for the alloys 1, 2, 3 and 4, respectively. These results suggest that the individual Mn and combined Mn, Ni additions may slightly increase grain size, while the combined Mn, Cr addition obviously increases the grain sizes of the as-cast alloy.

3.2. As-homogenized microstructure

Fig. 3 shows BSE images of the four alloys after homogenization with a rapid heating rate. For the base alloy 1, almost all the dark primary Mg_2Si phase has dissolved in the matrix, the fraction of primary α -AlFeSi phases has largely reduced, and the script-like phases have been broken up into spherical or elongated particles, as shown in Fig. 3(a and b). As alloy 1 is free of TMs, no dispersoids are observed in this alloy. For the Mn-added alloy 2, in addition to the dissolution of the primary phases, a

large number of dispersoids with rod-like or rectangular morphology and sizes of 100–200 nm have formed. The distribution of dispersoids is rather inhomogeneous: DFZs can be observed in the grain interior as well as close to grain boundaries, and a very low density of coarse dispersoids is formed in the DFZ of the grain interior. This phenomenon is mainly caused by the macrosegregation of Mn during casting. The inhomogeneous dispersoid distribution is detrimental to the alloy mechanical properties. For alloys 3 and 4, similarly inhomogeneous dispersoid distributions can be observed as well. As shown in Table 3, the average dispersoid sizes in the three alloys are measured as $165 \pm 5 \text{ nm}$, $257 \pm 23 \text{ nm}$ and $166 \pm 6 \text{ nm}$, respectively. This result suggests that the combined addition of Mn and Cr is not effective in reducing individual dispersoid sizes compared with individual Mn and combined Mn, Ni addition for the case of homogenization with a rapid heating rate.

Fig. 4 shows BSE images of the four alloys after homogenization with a slow heating rate. Compared with the rapidly homogenized sample, no clear difference is observed for the base alloy 1. For the three TM-added alloys, however, it can be seen that the dispersoids are now uniformly distributed in the Al matrix. The DFZ in the grain interior has almost disappeared, and the DFZ close to grain boundaries has significantly decreased in width. The measured DFZ areas for the three alloys under the two homogenization heating rates are shown in Fig. 5. The BSE-SEM images with the DFZs highlighted are shown in supplementary materials Fig. S2. It is clear that reducing the homogenization heating rate can appreciably reduce the sizes of the DFZs. Alloy 4 has a higher fraction of DFZ compared with alloys 2 and 3. In addition to the changes observed for the DFZs, the dispersoid sizes are generally considerably reduced and the number densities increased, compared to the outcome when homogenizing with a rapid heating rate. These results together suggest that reducing the heating rate of homogenization can clearly promote a uniform distribution of fine dispersoids. The average dispersoid sizes obtained with a slow heating rate for alloys 2–4 are measured as: $131 \pm 4 \text{ nm}$, $138 \pm 5 \text{ nm}$ and $134 \pm 4 \text{ nm}$, respectively (see Table 3). The three TM-added alloys show similar dispersoid size, implying that individual versus combined addition of TMs has no strong effect on this parameter.

To reveal the structure and composition of the dispersoids formed in the three TM-added alloys, bright-field TEM (BF-TEM) characterization was performed for these alloys after homogenization with a slow heating rate. Fig. 6(a–c) show BF-TEM images of the dispersoids formed under this condition. Alloy 3 with combined Mn, Cr addition exhibits the densest dispersoid distribution among the three alloys. The average diameter and number density of the dispersoids formed in the three TM-

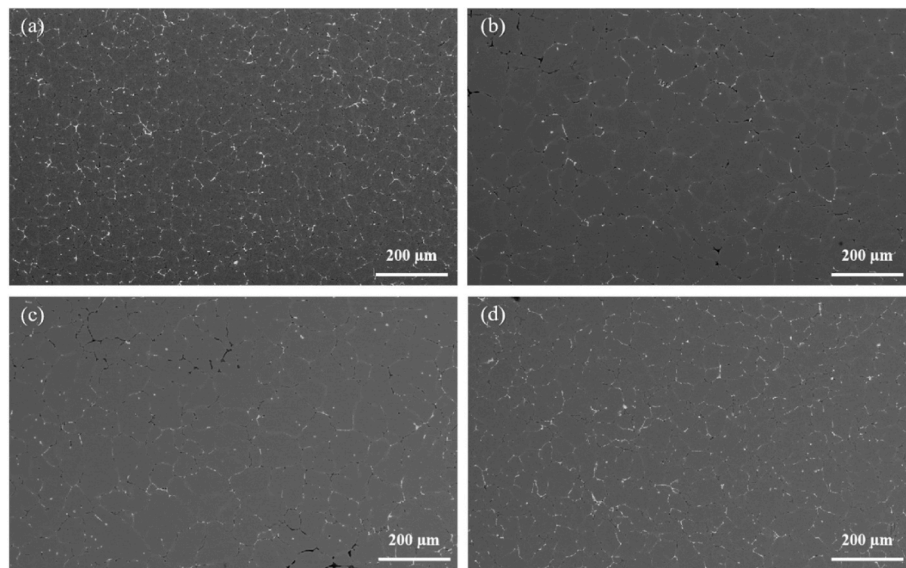


Fig. 1. BSE images of the four examined alloys in as-cast condition. (a) alloy 1, (b) alloy 2, (c) alloy 3, (d) alloy 4.

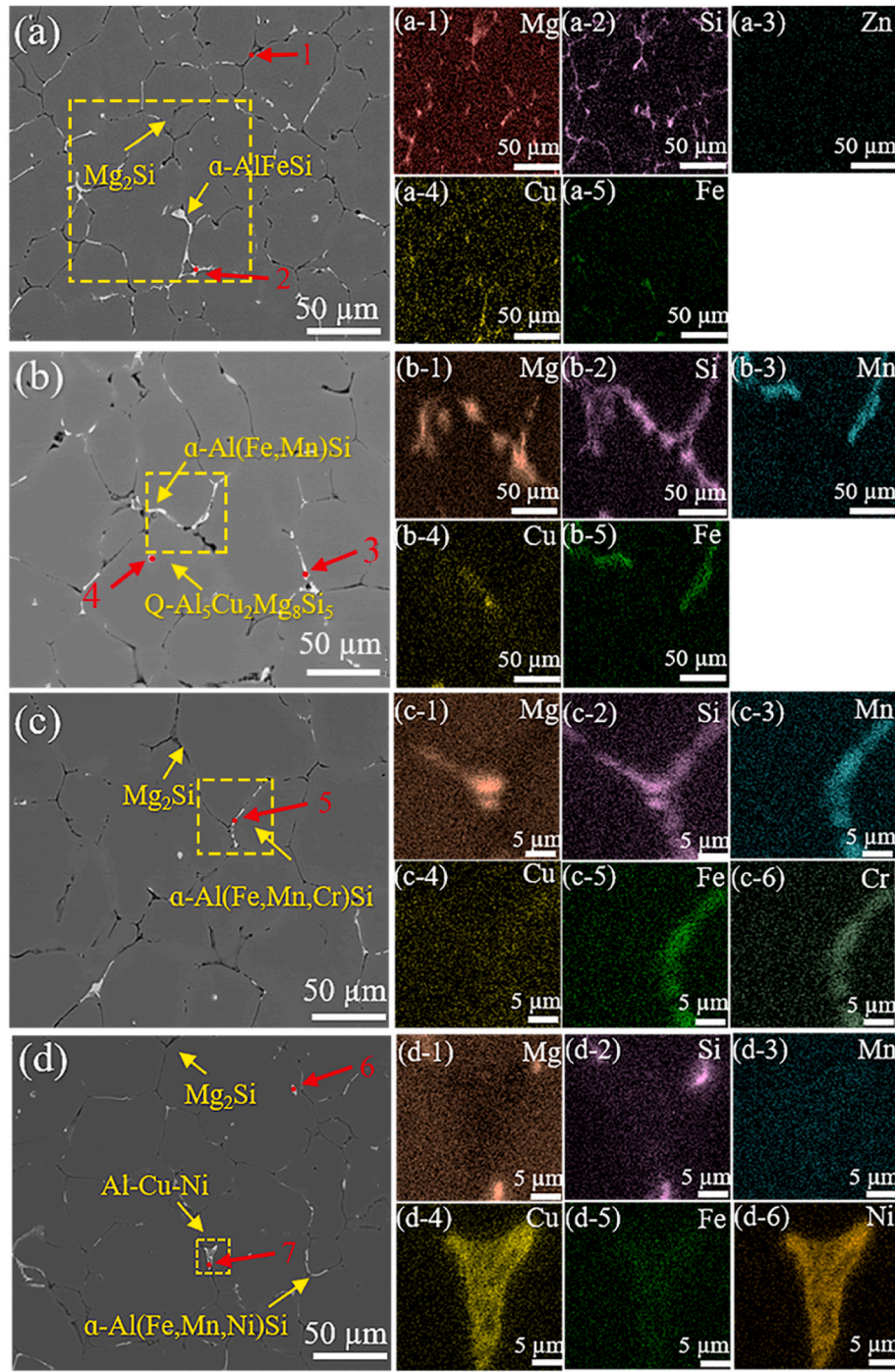


Fig. 2. Enlarged BSE images and EDX mappings of the primary phases formed in the four as-cast alloys. (a) alloy 1, (b) alloy 2, (c) alloy 3, (d) alloy 4. The particles for EDS point analysis are marked by red arrow, but area for EDS mapping is marked by yellow dashed squares. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Chemical composition (at. %) of primary phases marked in Fig.2.

	Al	Mg	Si	Fe	Mn	Cr	Ni	Cu
Point 1	52.57	32.14	13.96	0.25				0.11
Point 2	87.60	1.76	4.98	5.54				0.08
Point 3	78.42	1.59	8.28	3.60	7.75			0.34
Point 4	93.27	2.31	3.10	0.21	0.53			0.50
Point 5	76.85	1.58	6.37	3.86	9.05	1.69		0.57
Point 6	80.53	1.36	7.05	2.66	5.80		1.67	0.84
Point 7	82.60	1.58	0.94	0.28	0.21		8.40	6.17

added alloys were quantified based on the TEM images, with the results shown in Table 4. It can be seen that the average dispersoid size quantified by TEM is generally smaller than the value quantified by SEM (see Table 3). This difference is mainly due to some small dispersoids escaping detection by SEM due to the limited resolution of this technique. In agreement with the SEM results, the three TM-added alloys show similar dispersoid size according to the TEM-based analysis. The number density of the dispersoids formed in alloy 3 is highest, while it is lowest in alloy 4. This result indicates that the combined Mn, Cr addition can promote dispersoid formation, while combined Mn, Ni addition does not facilitate such formation.

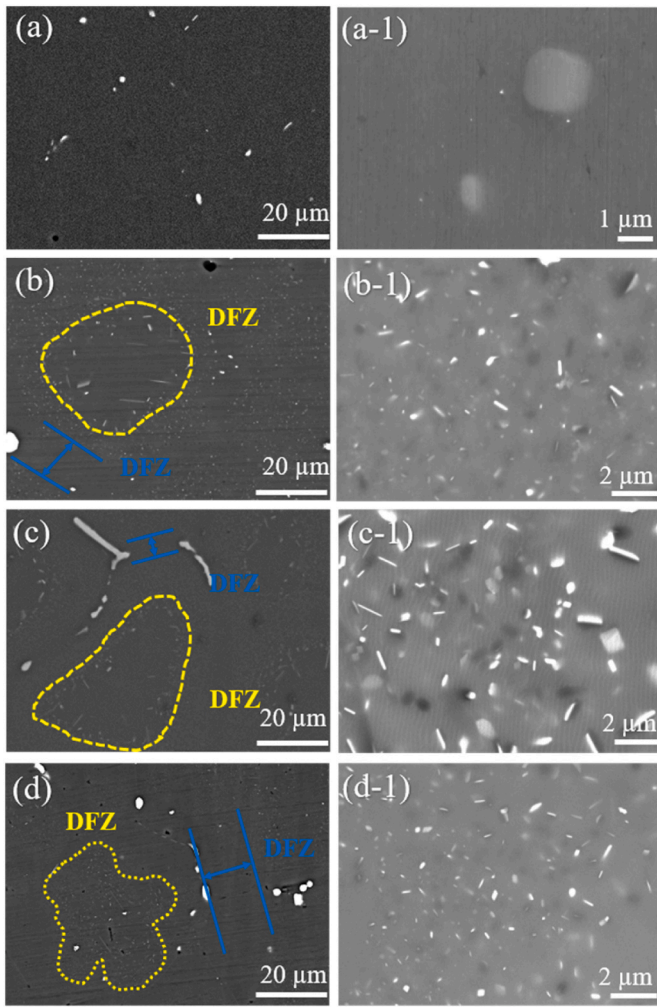


Fig. 3. BSE-SEM images of dispersoids formed in the four examined alloy following homogenization with a rapid heating rate: (a) alloy 1 (no dispersoids), (b) alloy 2, (c) alloy 3, (d) alloy 4. (a-1), (b-1), (c-1), (d-1) are enlarged images of regions in (a), (b), (c) and (d), respectively.

Table 3

Average diameter of the dispersoids formed in the three TM-added alloys, as quantified from SEM images.

Alloy	Average diameter (nm)
Alloy 2-Rapid heating	165 ± 5
Alloy 3-Rapid heating	257 ± 23
Alloy 4-Rapid heating	166 ± 6
Alloy 2- Slow heating	131 ± 4
Alloy 3- Slow heating	138 ± 5
Alloy 4- Slow heating	134 ± 4

High-resolution TEM (HRTEM) images and corresponding fast Fourier transform (FFT) patterns of the dispersoids formed in alloy 2–4 are shown in Fig. 6(d and e), respectively. It is clear that the dispersoids formed in the three alloys exhibit the same face-centered cubic (fcc) structure with the lattice parameter as being in the range 1.26–1.27 nm, corresponding well with previous work [15]. The EDX mappings of the dispersoids formed in the three alloys are shown in Fig. 7 and the measured compositions of different dispersoids are shown in Table 5. In alloy 2, the dispersoid can be clearly identified as α -Al(Fe, Mn)Si phase with an average composition of 81.2Al-7.4Si-0.4Mg-0.8Cu-10.1Mn-0.8Fe (at.%). In alloy 3, about 4.4 at.% Cr is incorporated in the dispersoid, resulting in an α -Al(Fe,Mn,Cr)Si phase with an average

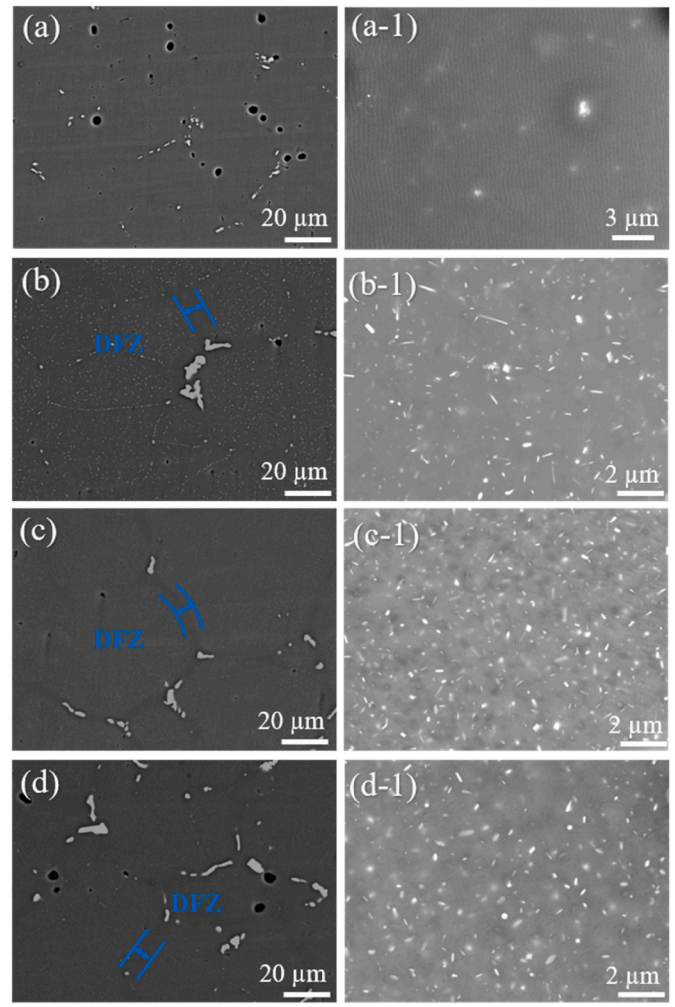


Fig. 4. BSE-SEM images of dispersoids formed in the four examined alloys following homogenization with a slow heating rate: (a) alloy 1 (no dispersoids), (b) alloy 2, (c) alloy 3, (d) alloy 4. (a-1), (b-1), (c-1), (d-1) are enlarged images of regions in (a), (b), (c) and (d), respectively.

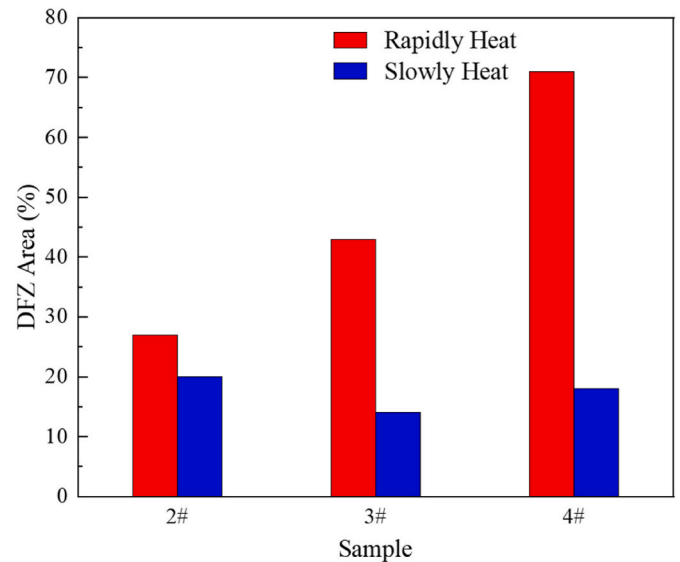


Fig. 5. Measured area fractions of the DFZs formed in the three TM-added alloys for the two heating rates applied in experiment.

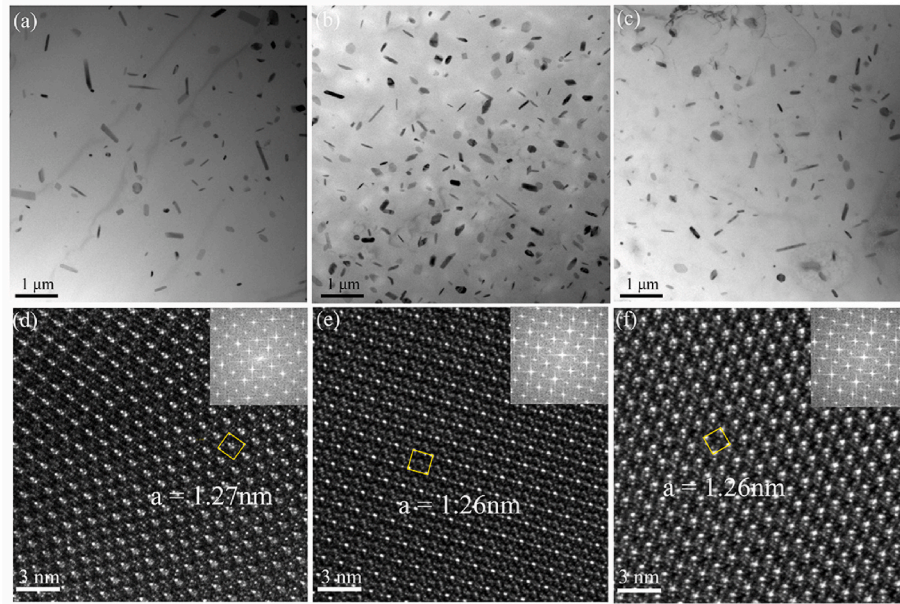


Fig. 6. (a, b, c) BF-TEM images of the dispersoids formed in the three TM-added alloys after homogenization with a slow heating rate. (d, e, f) HRTEM images of the dispersoids from (a, b, c), viewed along the $\langle 100 \rangle_{\text{Al}}$ zone axis. The unit cell of the dispersoid is marked by yellow square. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 4

Average diameter, number density and volume fraction of the dispersoids formed in the three TM-added alloys as quantified by TEM images.

Alloy	Average diameter (nm)	Number density (μm^{-3})	volume fraction (%)
Alloy 2	113 ± 10	19.12 ± 5.18	1.39 ± 0.58
Alloy 3	125 ± 9	29.65 ± 16.90	2.00 ± 0.22
Alloy 4	118 ± 10	15.63 ± 3.00	1.18 ± 0.24

composition of 70.1Al-13.5Si-1.6Cu-8.8Mn-4.9Cr-0.4Fe (at.%). In alloy 4, only minor Ni levels (0.5 at.%) are incorporated in the dispersoid, leading to an $\alpha\text{-Al(Fe,Mn,Ni)Si}$ phase with an average composition of 79.5Al-8.1Si-1.2Cu-10.3Mn-0.5Ni-0.6Fe (at.%). Compared with the coarse $\alpha\text{-Al(Fe,Mn,Cr)Si}$ primary phase of Fig. 8, hosting 3.9 at.% Fe, the Fe concentration in the $\alpha\text{-Al(Fe,Mn)Si}$ dispersoid is much lower. The above results indicate that although the dispersoids formed in the three alloys have different compositions (different concentrations of Cr or Ni), the structure and lattice parameter of the dispersoids are not changed. In other words, the observed incorporation of Cr and Ni in the Al(Mn, Fe)Si dispersoids does not affect the particle lattice parameter.

3.3. Age hardening and final mechanical properties

Due to the uniform distribution of dispersoids formed in the homogenized samples with slow heating rate, the slow heating homogenized samples are selected for the hot- and cold-rolling, followed by solution treatment and immediate artificial aging at 180 °C for different durations. Fig. 9(a) shows artificial aging hardness curves of the four alloys. An enlarged view of the hardness curves, marked by the orange dashed rectangle of Fig. 9(a), is shown in Fig. 9(b). All the four alloys display similar profiles: the hardness rapidly increases to the maximum value, followed by an over-aging stage with decreasing hardness. It can be seen that alloy 3 exhibits the highest peak-aging hardness. Alloys 2 and 4 show similar peak-aging hardness, while the base alloy 1 has the lowest value. It should be noted that these differences are very small, the peak hardness of alloy 3 only being 6HV higher than that of alloy 1. The hardness increments ΔHV for the four alloys during peak-aging are shown in Fig. 9(c). Alloy 1 shows the highest ΔHV of 83.5HV, while the

TM-added alloys have slightly lower ΔHV values (79.1, 80.4 and 79.4 for alloy 2, 3 and 4, respectively). This result implies that although Mn, Cr or Ni addition can increase the peak hardness, the age hardening responses of the alloys are actually slightly decreased. This is mainly due to Si atoms being consumed during the formation of $\alpha\text{-Al(Fe,TM)Si}$ phase, with less Si being available for the precipitation of metastable (Al-Mg-Si-Cu) hardening phases. The evolution in the electrical conductivities (ρ) for the four alloys during aging is shown in Fig. 9(d). The base alloy 1 shows an overall higher electrical conductivity compared with the TM-added alloys. This is mainly due to the TM atoms (Mn, Cr, Ni) in solid solution greatly decreasing this quantity. The $\Delta\rho$ of the base alloy 1 is also higher than the TM-added alloys, which further confirms the higher age hardening response of the base alloy 1.

To reveal the influence of the TMs on the final microstructures of the alloys, the grain size and second phase distributions of each alloy were characterized (see Figs. 10 and 11). As the four alloys do not exhibit a clear difference in age hardening response, the nano-sized Al-Mg-Si-Cu metastable hardening precipitates are not studied in the present work. Fig. 10 shows the inverse pole figure (IPF) and image quality (IQ) map, superimposed with grain boundary (GB) maps, of the four alloys. The base alloy 1 has equiaxed grains, while the three TM-added alloys exhibit slightly elongated grains parallel to the rolling direction (RD). Most of the grains in the four alloys are recrystallized, although a few low-angle grain boundaries (LAGBs) may be observed. From the grain size distributions of the aged samples shown in Fig. 10(i-l), the average grain sizes of the four alloys are measured as: 70.62 μm , 19.32 μm , 21.54 μm and 24.03 μm , respectively. It is clear that the TM-added alloys have smaller grain sizes compared with the base alloy 1, while no clear grain size difference can be detected among the latter three alloys. The misorientation angle distributions of the four alloys are shown in Fig. 10 (m-p). The three TM-added alloys show a similar fraction of LAGBs compared with the base alloy 1. These results indicate that although the presence of fine dispersoids cannot effectively suppress the recrystallization upon solution treatment at 560 °C, these dispersoids can still promote grain refinement. Fig. 11 shows BSE images of the dispersoids formed in the four peak-aged alloys. Compared with the dispersoids formed during homogenization, the dispersoid distribution is even more uniform, and most of the dispersoids have a spherical or near-spherical

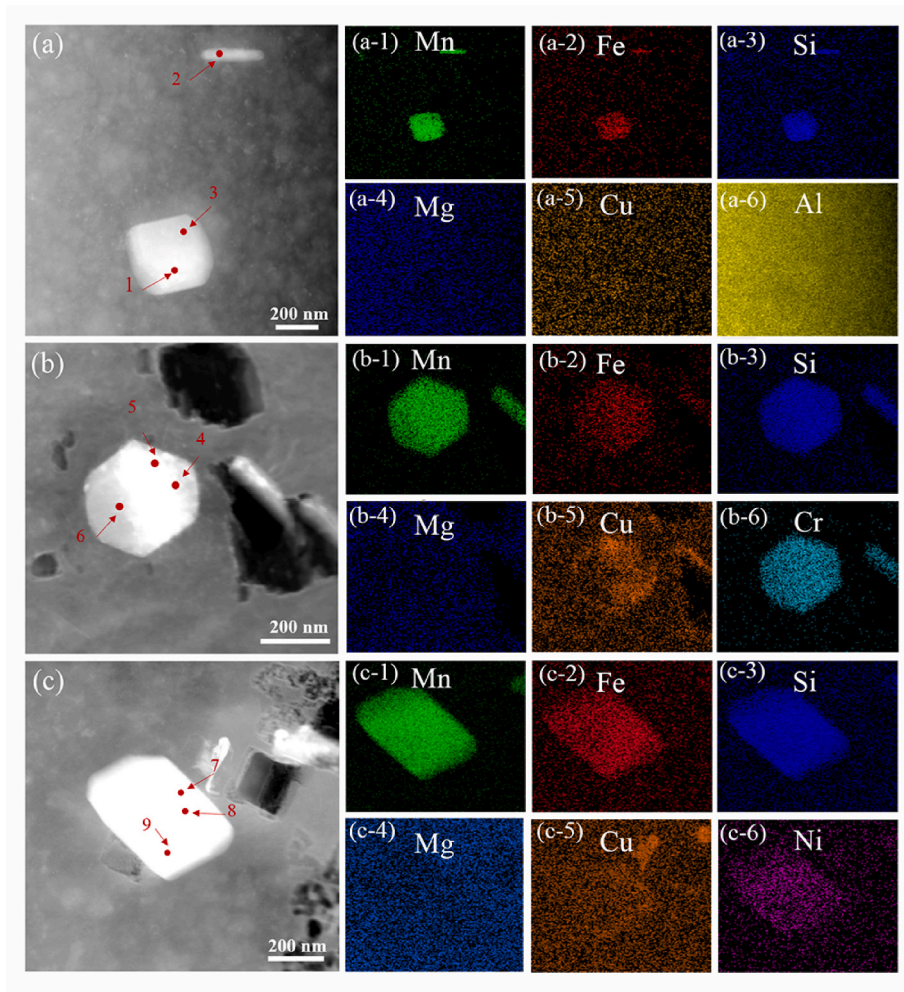


Fig. 7. (a, b, c) HAADF-STEM images and corresponding EDX mappings of the dispersoids formed in the three TM-added alloys.

Table 5

The chemical composition of the dispersoids and embedding matrix in the vicinity of the points highlighted in Fig. 7 (at.%).

Points	Al	Si	Mg	Cu	Mn	Cr	Ni	Fe	Suggested phase
1	80.1	7.7	0.2	1.1	10.1	–	–	0.8	Al(Fe,Mn)Si
2	83.1	7.3	0.6	0.8	7.5	–	–	0.7	Al(Fe,Mn)Si
3	80.5	7.4	0.3	0.6	10.1	–	–	1.1	Al(Fe,Mn)Si
4	70.5	13.3	–	1.6	9.4	4.7	–	0.4	Al(Fe,Mn,Cr)Si
5	69.0	13.5	–	3.4	8.5	5.2	–	0.4	Al(Fe,Mn,Cr)Si
6	70.9	13.7	0.1	1.2	8.7	4.9	–	0.4	Al(Fe,Mn,Cr)Si
7	79.5	8.1	1.3	0.7	10.3	–	0.5	0.8	Al(Fe,Mn,Ni)Si
8	80.8	8.8	1.1	0.6	10.2	–	0.4	0.3	Al(Fe,Mn,Ni)Si
9	79.3	8.3	1.0	0.7	10.6	–	0.6	0.7	Al(Fe,Mn,Ni)Si

morphology. The dispersoid size and number density have not changed compared to the values obtained upon homogenization.

Fig. 12 shows TEM bright-field images of the nano-sized precipitates formed in the four peak-aged alloys. It is clear that high densities of precipitates are uniformly distributed for the four alloys.

HAADF-STEM images of Fig. 13 show that all the precipitates exhibit a disordered structure, and two types of precipitates can be identified. The precipitate presented in Fig. 13(a) can be identified as disordered β'' phase, some Cu atoms are incorporate in the precipitates and the substructure of β'' -“low density cylinder” (LDC, marked by green dashed circles) is randomly distributed in the precipitate, leading to disordered structure of the precipitates. No β'' unit cell is observed. The precipitates shown in Fig. 13(b) is identified as QP2 phase (also termed as L phase) containing the unit cell of the C phase. The formation of the disordered

β'' and QP2 phase is mainly due to the combination of Mg-rich and high Cu composition as suggested in our previous work [3]. Statistical number density and cross-section of precipitates of the four alloys are shown in Table 6. The three TM-added alloys have similar precipitate distribution, while the precipitates formed in alloy 1 has a smaller size and higher number density compared with the three TM-added alloy, corresponding well the higher age hardening response of the alloy 1. Fig. 14 show the distribution of the α -Al(Fe,TM)Si dispersoids and the nano-sized precipitates formed in the alloy 2, no heterogeneous precipitation of the β'' phase is observed at the dispersoid/matrix interface, suggesting the heterogeneous precipitation is suppressed in this alloy due to the high quenching rate.

Fig. 15(a) shows the tensile true stress-true strain curves of the four alloys after peak-aging (3h for alloy 3, 5h for alloy 1 and so on), with the

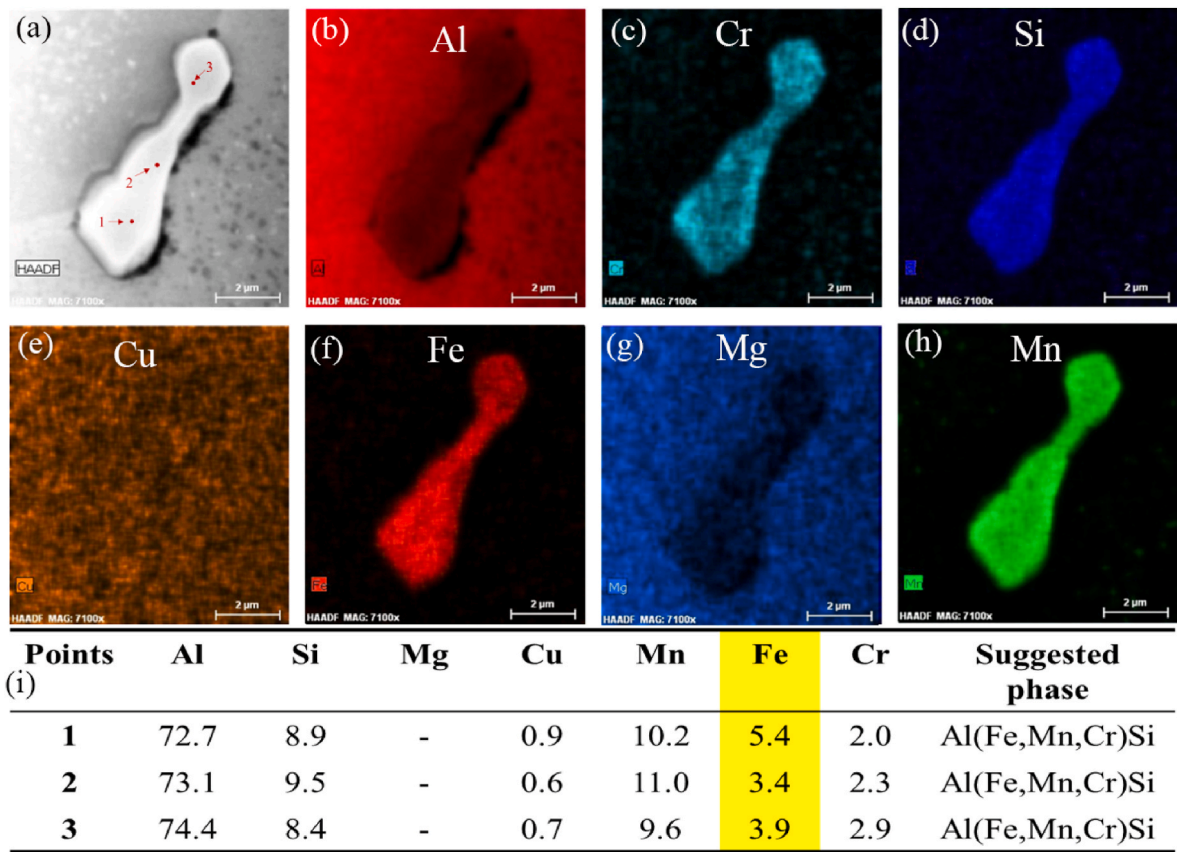


Fig. 8. HAADF-STEM image and corresponding EDX mapping of the Al(Mn,Fe,Cr)Si primary phase formed in alloy 3.

corresponding tensile properties presented in Table 7. It can be seen that the base alloy has both the lowest ultimate tensile strength (UTS) and elongation (EL) (391.2 ± 8.3 MPa and 6.6 ± 2.2 %, respectively). In other words, the addition of Mn, Cr or Ni can simultaneously increase the alloy strength and elongation. Among the three TM-added alloys, the Mn-added alloy 2 has the maximum strength combined with high elongation. The UTS of this alloy reaches 424.4 ± 5.7 MPa and the elongation is doubled compared with alloy 1. Alloy 3 shows slightly lower strength compared with alloy 2, but its elongation (18.2 ± 3.6) is the highest among the three alloys, almost three times the value for alloy 1. This alloy exhibits an excellent combination of strength-elongation, better than most of the reported 6xxx series alloys (see Fig. 15(b)) [23–37]. Alloy 4 has a slightly lower strength and decreased elongation compared with alloy 2, indicating that the combined addition of Mn, Ni does not cause an improvement of the mechanical properties compared with individual Mn addition. Therefore, alloy 3 with combined Mn, Cr addition is selected as the best alloy among three TM-added alloys as it provides simultaneous improvement of strength and elongation.

To compare the fracture behavior of the investigated alloys, the fractography of the four peak-aged alloys are shown in Fig. 16. The base alloy 1 shows a completely intergranular fracture mode; obvious river-type cleavage planes with a very low density of dimples are present in Fig. 16(a), corresponding well with the low elongation of this alloy. By comparison, the three TM-added alloys (Fig. 16(b–d)) present transgranular fracture surfaces with high densities of dimples. Alloy 3 has the highest density of dimples among the three TM-added alloys, which is in good agreement with the observation of a superior elongation for this alloy (Fig. 15(a)). This result confirms that addition of TMs can transform the fracture mode from intergranular to transgranular fracture.

4. Discussion

4.1. The effects of TMs on the precipitation of dispersoids during homogenization

It is well known that Mn addition can promote the transformation of the needle-like β -AlFeSi phase to Chinese-script α -Al(Fe,Mn)Si primary phase and thus reduce the detrimental effect of Fe. In addition, the precipitation of large numbers of α -Al(Fe,Mn)Si dispersoids during homogenization can suppress the recrystallization and thus improve the properties of the Al–Mg–Si alloys [38,39]. In the present work, due to the rather low alloy Fe content, the β -AlFeSi phase is not formed even in the base (TM-free) alloy. The incorporation of TMs in the α -Al(Fe,TM)Si primary phases do not change the morphology and size of the primary phases that do form. The grain sizes of the as-cast alloys increased after TM addition, especially for the case of combined Mn, Cr addition. This is mainly due to the Mn, Cr addition decreasing the solidification temperature of the alloy, which in turn increases the superheat degree of the melt and extends the solidification time [40].

For the precipitation of the dispersoids during homogenization, the various individual and combined additions of TMs examined are found to have significantly different influence. For the Mn-added alloy, owing to the high solubility of Mn in fcc Al and its relatively fast diffusion rate [21], the Mn atoms readily precipitate and form a large number of Al (Mn, Fe)Si dispersoids. As the solubility of Fe in the Al matrix is extremely low [41], the Fe concentration in these dispersoids is rather limited (about 0.7 at.%). For the combined addition of Mn and Cr, Mn atoms trigger an initial nucleation of Al(Mn, Fe)Si dispersoids due to their high diffusion rate. Owing to the slow diffusion rate and the low solubility of Cr, the Cr atoms are gradually incorporated in these dispersoids, modifying them into the Al(Mn, Cr, Fe)Si phase. For the combined Mn, Cr addition, firstly, as the Cr atoms are replacing the Mn

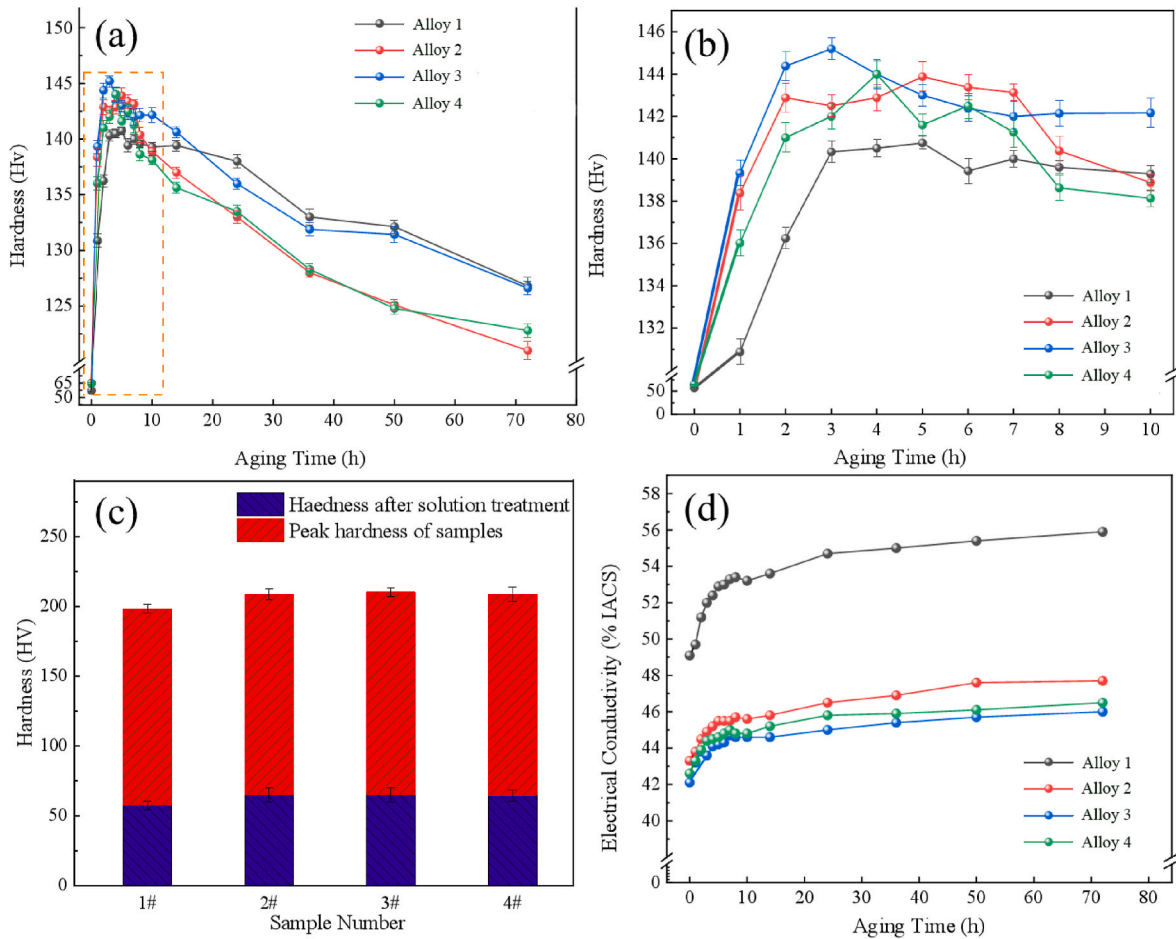


Fig. 9. (a) Artificial aging hardness curves of the four alloys at 180 °C, (b) enlarged view of the hardness curves (region marked by an orange dashed rectangle in (a)), (c) hardness increments for the four alloys during peak-aging, (d) The evolution of electrical conductivities (ρ) for the four alloys during aging at 180 °C. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

sites of the dispersoids, the number density of the dispersoids is thus determined by the (Mn + Cr) content. Therefore, the high content of the (Mn, Cr) in alloy 3 can significantly increase the number density of the dispersoids. Besides, owing to the slow diffusion rate and the low solubility of Cr, the incorporation of Cr atoms in the Al(Fe, Mn, Cr)Si dispersoids can slow down the coarsening rate of these dispersoids, which further increase the number density of the dispersoids [42]. For the combined addition of Mn and Ni, due to the rather low solubility of Ni in the Al matrix and a strong interaction between Cu and Ni, the majority of the Ni atoms are incorporated in the coarse AlCuNi intermetallic phase [43]. The AlCuNi phase will consume high concentrations of the Cu and Ni atoms present in the alloy, and thus reduce the hardening effect of the alloy. The AlCuNi phase is also detrimental to the alloy ductility due to its brittle nature. Only a low concentration of Ni atoms (about 0.5 at.%) can be incorporated in the dispersoid to form the Al(Mn, Ni, Fe)Si phase. As a consequence, the effect of Ni on the precipitation of the Al(Mn, Ni, Fe)Si dispersoid is rather limited, and the precipitation of Al(Mn, Ni, Fe)Si dispersoids is even slightly reduced compared with the case of the individual Mn added alloy. From the above analysis, it can be concluded that the combined addition of Mn, Cr is the most effective in promoting the dispersoid precipitation, followed by individual Mn and combined Mn, Ni additions. Reducing the heat rate of the homogenization can significantly reduce the individual size and increase the number density of the dispersoids. Also, the level of spatial homogeneity in the dispersoid distribution is obviously improved by this approach. As it is generally known that the precipitation temperature of the Al(Fe, Mn)Si dispersoids is about 350 °C [44], decreasing the heating rate of the

homogenization can increase the nucleation duration of the dispersoids, resulting in a finer ultimate distribution. Besides, the slow heating rate can also facilitate the diffusion of the Mn, Cr, which reduces the inhomogeneous distribution of the dispersoids. At the end of the slow heating rate procedure, the DFZ in the grain interior has disappeared and the DFZ along the grain boundaries has been markedly reduced.

4.2. The effect of dispersoids on the recrystallization and mechanical properties

After homogenization, the samples are hot-rolled, cold-rolled, solution treated and artificially aged to obtain peak hardness. The nano-sized precipitates contribute the largest hardening response for the four alloys. Due to the consuming of Si atoms during the formation of Al (TM, Fe)Si dispersoids, the base alloy 1 has a slight higher age hardening response than the TM-added alloys, while the three TM-added alloys exhibit similar age hardening response. Therefore, the main reason responsible for the property differences of these alloys is dispersoids formed during homogenization. The presence of different number densities of dispersoids can significantly affect the final microstructure and mechanical properties of the different TM-added alloys. The main effects of dispersoids on the alloy mechanical properties are as follows: (i) suppressing the recrystallization upon solution treatment, (ii) providing Orowan hardening, and (iii) reducing stress concentration on grain boundaries. As for point (i), many studies have reported that the formation of a high number density of dispersoids produces an increased Zener drag, improving recrystallization resistance during subsequent

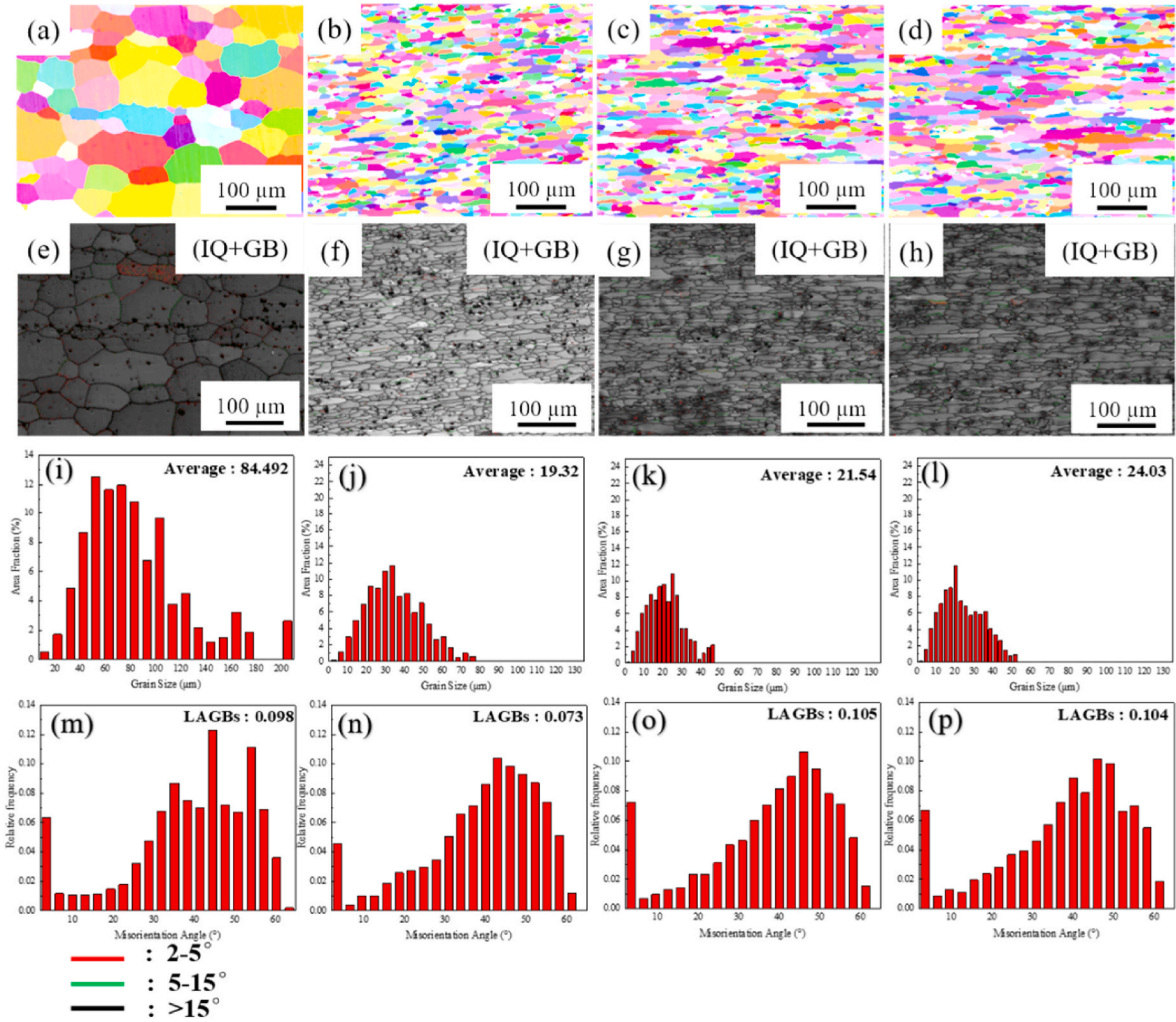


Fig. 10. (a–d) IPF (viewed along transverse direction (TD)), (e–h) IQ map, superimposed with GB maps, (i–l) grain size distributions, and (m–p) misorientation angle distributions for alloys 1–4, respectively.

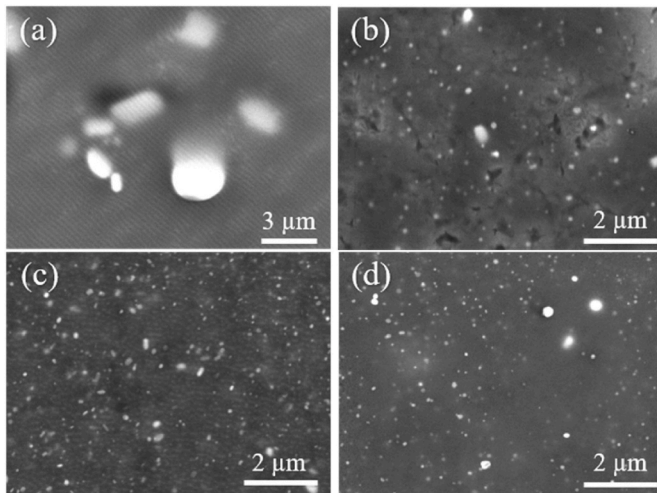


Fig. 11. BSE images of the dispersoids formed in the four peak-aged alloys. (a) alloy 1, (b) alloy 2, (c) alloy 3, (d) alloy 4.

annealing treatment [12,16]. In the present work, although clear recrystallization has occurred for the four alloys, the presence of dispersoids in the TM-added alloys has evidently obstructed the migration

of grain boundaries, as revealed by a significant reduction of the grain sizes. Among the three TM-added alloys, alloy 2 and 3 have the smallest grain sizes, while alloy 4 exhibits slightly larger grain sizes. These results suggest that the different dispersoid distributions in the three TM-added alloys do not show clear differences in their ability to refine the alloy grains.

As for point (ii), for the Orowan hardening provided by the dispersoids, the dispersoid-induced yield strength increment is generally predicted using the Orowan bypass equation [21,45]:

$$\Delta\sigma = \frac{0.84MGb}{2\pi(1-\nu)^{1/2}\lambda} \ln \frac{r}{b} \quad (1)$$

$$\lambda = r \left(\frac{2\pi}{3f} \right)^{1/2} \quad (2)$$

In these expressions, M is the Taylor factor of aluminum (3.06), G is the shear modulus of fcc Al (25.4 GPa), b is the Al Burgers vector (0.286 nm), ν is Poisson's ratio for Al (0.34), λ is the dispersoid interparticle spacing, f is the dispersoid volume fraction and r is the dispersoid average radius. The calculated dispersoid-induced contributions $\Delta\sigma$ to the yield strength are presented in Table 8. It can be seen the Orowan hardening caused by dispersoids is small. The three TM-added alloys produce very similar values of $\Delta\sigma$, corresponding well with their similar YS shown in Fig. 15. The TM-added alloys have higher strength than the

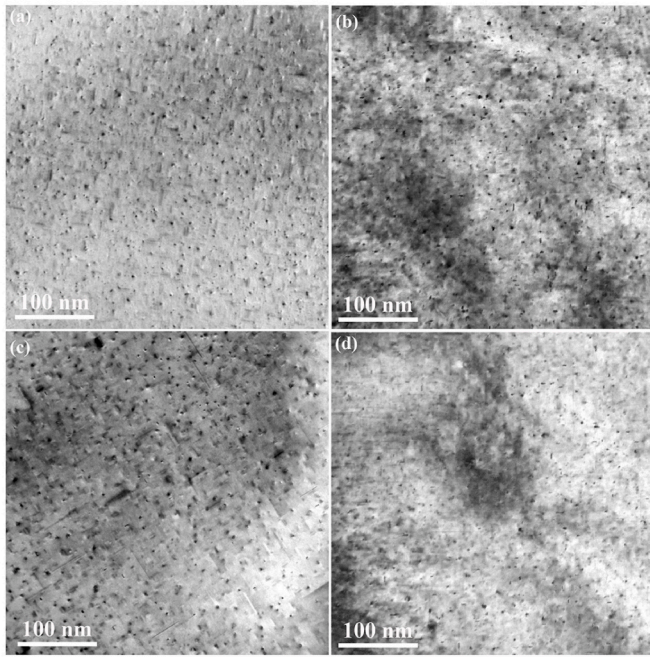


Fig. 12. TEM bright-field images of the four alloy under peak aged condition. (a) alloy 1, (b) alloy 2, (c) alloy 3, (d) alloy 4.

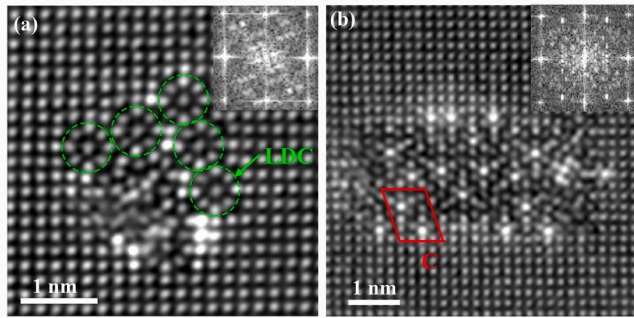


Fig. 13. HAADF-STEM images of the precipitates formed in the peak aged alloy 2, (a) disordered β phase, (b) disordered QP phase.

Table 6

Statistical number density and cross-section of precipitates of the four peak-aged alloys.

	Cross section (nm^2)	Number density ($\times 10^{-5}/\text{nm}^3$)
Alloy 1	2.6	4.76
Alloy 2	3.1	4.92
Alloy 3	2.9	4.91
Alloy 4	3.05	4.84

base alloy 1 due to the high dispersoid number density.

As for point (iii), the presence of fine dispersoids can also affect the plastic deformation behavior of the Al–Mg–Si alloys. The fracture mode changes from intergranular fracture to transgranular fracture after the addition of TMs. Poole et al. [14] revealed that the plastic strain was organized into slip bands, and the magnitude of the local shear strain decreased with reduced dispersoid spacing, resulting in lower stress concentration on the grain boundary and thus decreasing the propensity for intergranular fracture. Zhao et al. [46] reported the presence of high density of fine dispersoids strengthen the materials significantly, but their strengthening effect diminishes as strain increase, the internal stress due

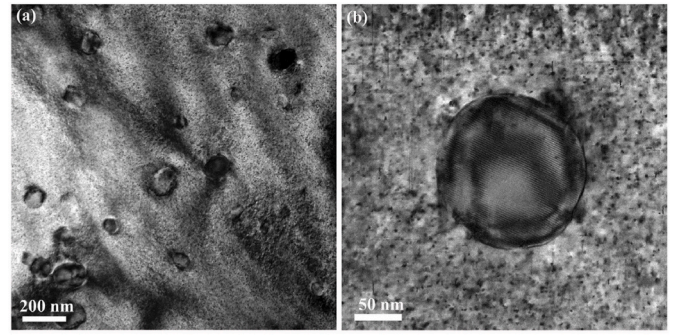


Fig. 14. (a) TEM bright-field image of the distribution of the α -Al(Fe,Mn)Si dispersoids and the nano-sized precipitates formed in the alloy 2, (b) enlarged image of (a).

to particles makes a contribution to strain hardening during the early deformation. Trink et al. [47] also reported that the Al–Mg–Si–Fe crossover alloy with high density of Fe-rich intermetallic phase exhibit high strain hardening due to the formation of geometrically necessary dislocations (GNDs) around the morphologically transformed Fe-rich primary phases. The surprising high elongation is achieved due to the combined effect of Fe-rich intermetallic refinement and their roundish morphology, refinement of grain sizes and more uniform deformation. Therefore, the formation of a high number density of dispersoids can produce uniformly deformation and thus significantly increase the alloy elongation. In the present work, as a result of the combined effect of Mn, Cr, the precipitation of the dispersoids for alloy 3 is strongly promoted. This alloy has the highest dispersoid density, resulting in a superior suppression of the stress concentration of grain boundary, which in turn produces a better strength-elongation combination than observed for most of the reported 6xxx aluminum alloys to date. In summary, the presence of fine dispersoids can not only moderately increase alloy strength, but also strongly increase alloy elongation. Compared with individual Mn addition and combined Mn, Ni addition, the combined addition of Mn, Cr is more effective in promoting the formation of Al (TM, Fe)Si dispersoids.

5. Conclusions

In the present work, the effect of individual and combined additions of different TMs (Mn, Cr, Ni) on the precipitation of dispersoids and the final mechanical properties of Al–Mg–Si–Cu alloys were systematically studied. The main conclusions are listed as follows.

- (1) In the as-cast alloys, Mg_2Si , Q and α -Al(Fe, TMs)Si are the main primary phases. After the addition of Mn, Cr, these elements can be incorporated into the α -AlFeSi phase and form the α -Al(Fe, Mn, Cr)Si primary phases, but the size and fraction of these phases are not changed. After the addition of Mn, Ni, only a minor concentration of Ni is incorporated into the α -AlFeSi primary phase, with most Ni atoms going into the AlCuNi eutectic phase.
- (2) During homogenization, the individual and combined additions of TMs are found to significantly affect the precipitation behavior of the dispersoids. The combined addition of Mn, Cr produces the highest number density of dispersoids among the three TM-added alloys. This is mainly due to the incorporation of Cr promoting dispersoid precipitation. Due to the low concentration of Ni in the dispersoids, the effect of Ni on the precipitation of Al(Mn, Ni, Fe) Si dispersoid is rather limited.
- (3) Decreasing the heating rate for the homogenization process can not only induce a higher number density of finer dispersoids; also, the notable inhomogeneity in the dispersoid distribution observed for the case of a rapid heating rate is significantly reduced.

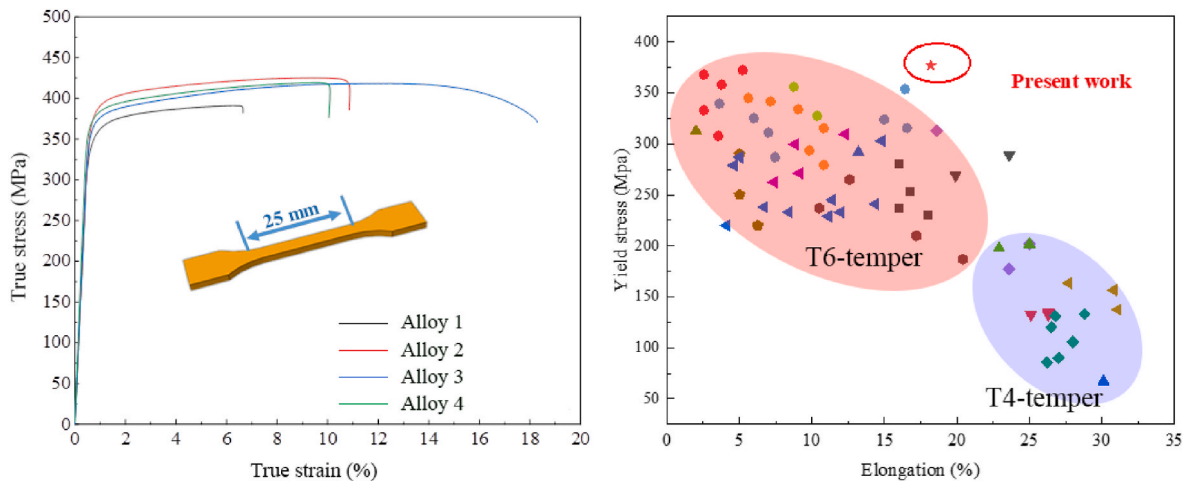


Fig. 15. (a) Tensile true stress-true strain curves of the four alloys after peak-aging, (b) the strength and elongation values for alloy 3, compared with results for other 6xxx alloys [23–37].

Table 7
Mechanical properties of four alloys.

Alloy	YS (MPa)	UTS (MPa)	Elongation (%)
Alloy 1	346.5 ± 10.1	391.2 ± 8.3	6.6 ± 2.2
Alloy 2	379.2 ± 5.5	424.4 ± 5.7	15.9 ± 4.7
Alloy 3	376.9 ± 1.3	421.9 ± 2.2	18.2 ± 3.6
Alloy 4	371.4 ± 8.2	419.4 ± 8.8	10.0 ± 2.0

Table 8
The calculated contribution of dispersoids to yield strength.

Alloy	Calculated $\Delta\sigma$ at 25 °C (MPa)
Alloy 2	16.59
Alloy 3	16.51
Alloy 4	14.95

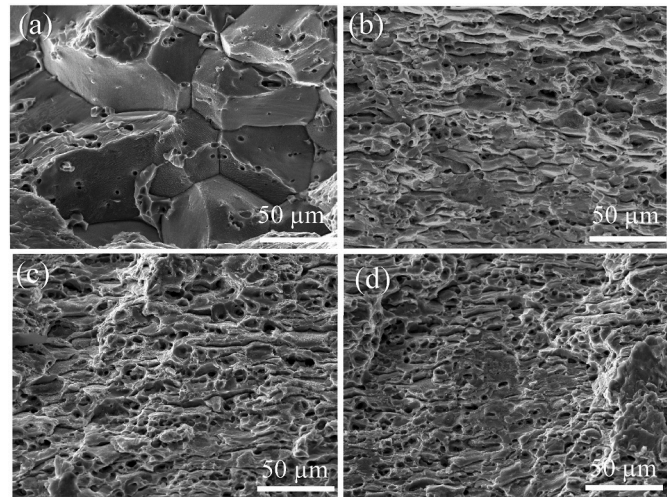


Fig. 16. SEM images of the tensile fracture surfaces for the four alloys after peak-aging. (a) alloy 1, (b) alloy 2, (c) alloy 3, (d) alloy 4.

(4) The presence of fine dispersoids can not only moderately increase the strength, but also strongly increase the elongation of the three TM-added alloys. For the combined Mn, Cr added alloy, the elongation is almost increased threefold compared with the base alloy. This is mainly due to the presence of the high number density of dispersoids causing grain refinement and reducing the stress concentration on the grain boundary.

CRediT authorship contribution statement

Lipeng Ding: Writing – original draft, Validation, Investigation, Funding acquisition. **Xuemei Zhang:** Investigation. **Chang Lu:** Investigation. **Chenglin Wang:** Supervision, Funding acquisition,

Conceptualization. **Yaoyao Weng:** Funding acquisition, Formal analysis. **Qipeng Dong:** Methodology. **Shaoqiang Xu:** Methodology. **Lingfei Cao:** Supervision, Funding acquisition, Conceptualization. **Zhihong Jia:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msea.2023.145962>.

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